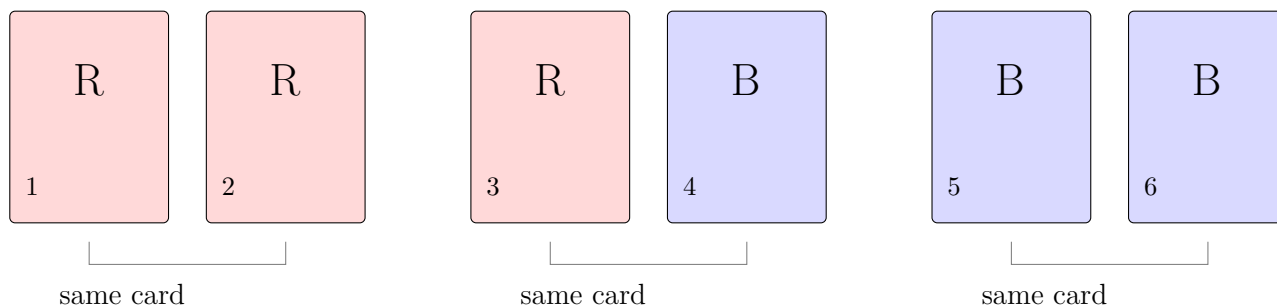
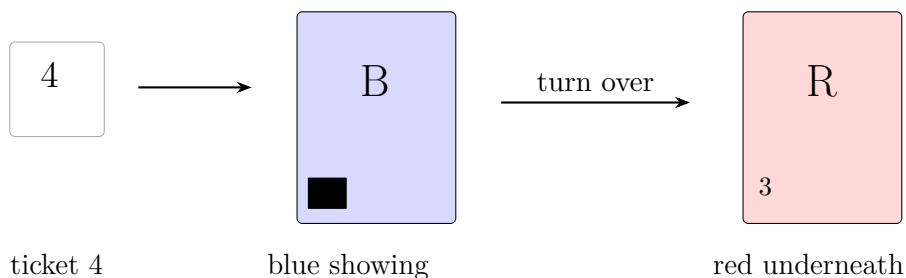


Each pair below is the two faces of one card. The small marks match the six tickets.



Use matching-size cards. Mix all six tickets and draw with equal chances. Behind a screen, choose the drawn face and set it upward. Keep the ticket secret and cover marks identically; reveal only the face color. Replace the ticket and mix each round.

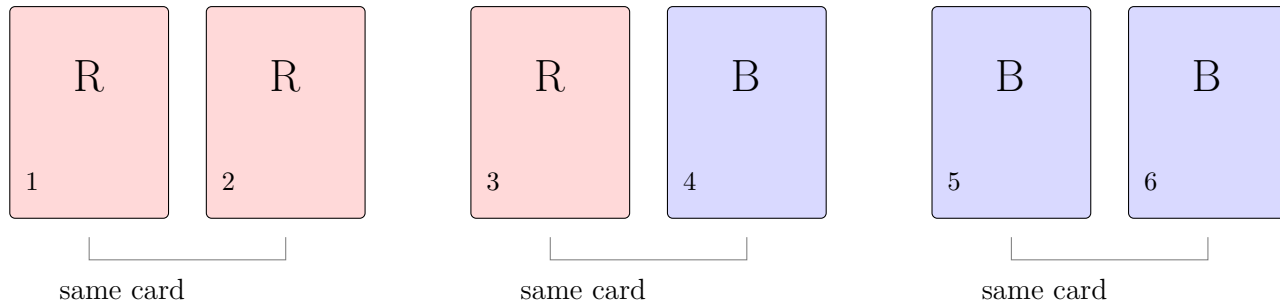


Problem 1: Given only a red showing color, determine the chances of the hidden colors exactly. Do the same for a blue showing color.

red showing

blue showing

Problem 2: A friend says, "Red is showing, so only the red/red and red/blue cards remain. Each is one card, so each must have the same chance." Does the selection rule support that argument?



Problem 3: If the chooser also reveals the face mark, what happens to the uncertainty? Explain why covering the mark matters to the original game.

Use only tickets 1 and 3, with equal chances. Choose and orient behind the screen, keep the ticket secret, and show its red face with the mark covered. Replace and mix each round.



Problem 4: Does this way of producing a red clue change the chance that red is hidden? Give an exact comparison with the six-face-ticket rule.

Problem 5: Could six red-showing rounds prove that the hidden-color chances differ? Give a possible set of hidden-color results that both chooser rules can produce.

Use a cup containing a subset of the original six tickets, with no repeats. Each selected ticket has the same chance.

Problem 6: Find every subset that makes red showing equally likely to hide red or blue. Your cup must sometimes show red; it may also show blue.

1	2	3	4	5	6
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Problem 7: Can you keep only whole cards and make that red-clue game fair, with red still possible? Keep one, two, or all three cards, and remove both tickets whenever a card goes.